Name: ______________________

Worksheet

Problem solving with transformations

Task A: Inverse problems

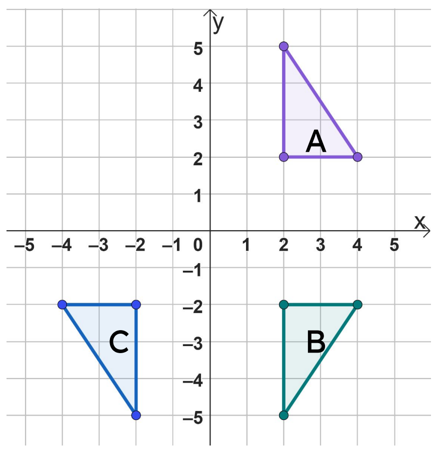
1) Shape M is translated by $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ to create shape N. What vector would translate shape N onto shape M?

2) Shape P is reflected in the x-axis to create shape Q. What transformation would map shape Q back on to shape P?

3) Shape R is enlarged by a scale factor of 4 from the origin. What enlargement scale factor would be needed to map back to shape R?

4) Shape A is reflected in the x-axis to become shape B. Shape B is then reflected in the y-axis to become shape C.

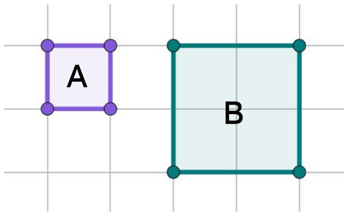
Fully describe a single transformation that can map shape C onto shape A.

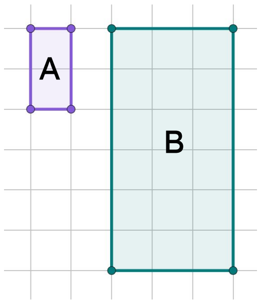


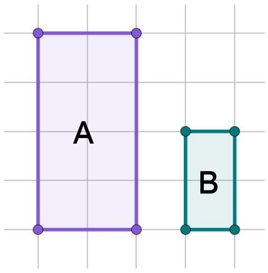
Task B: Perimeter and area

1) A shape has a perimeter of 10 cm.
   a) If it is translated its perimeter would be ___ cm.
   b) If it is rotated its perimeter would be ___ cm.
   c) If it is reflected its perimeter would be ___ cm.
   d) If it is enlarged by scale factor 0.5, its perimeter would be ___ cm.

2) Work out the perimeter and area of the object and image in the following enlargements. Use 1 square = 1 cm.

a)


b)


c)


Task C: Coordinate problems

1) The following points have been translated by $\begin{pmatrix} -5 \\ 3 \end{pmatrix}$
   Write the coordinate of the image.

   a) (a, b)

   b) $(a + 1, b + 3)$

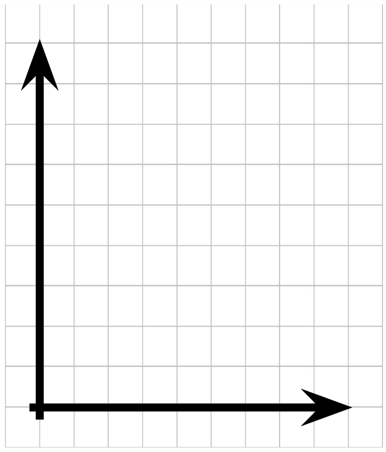
   c) $(a - 3, b + 3)$

   d) $(2a, 7 - b)$

   e) $(4 + a, b + 2)$

2) If these are the coordinates for the vertices, what transformation has occurred?
   You may wish to plot the coordinates.

   a) Object: A(1,4) B(1,1) C(2,1)
      Image: A′(4,4) B′(4,1) C′(5,1)



   b) Object: A(1,4) B(1,1) C(2,1)
      Image: A′(1,4) B′(1,7) C′(2,7)